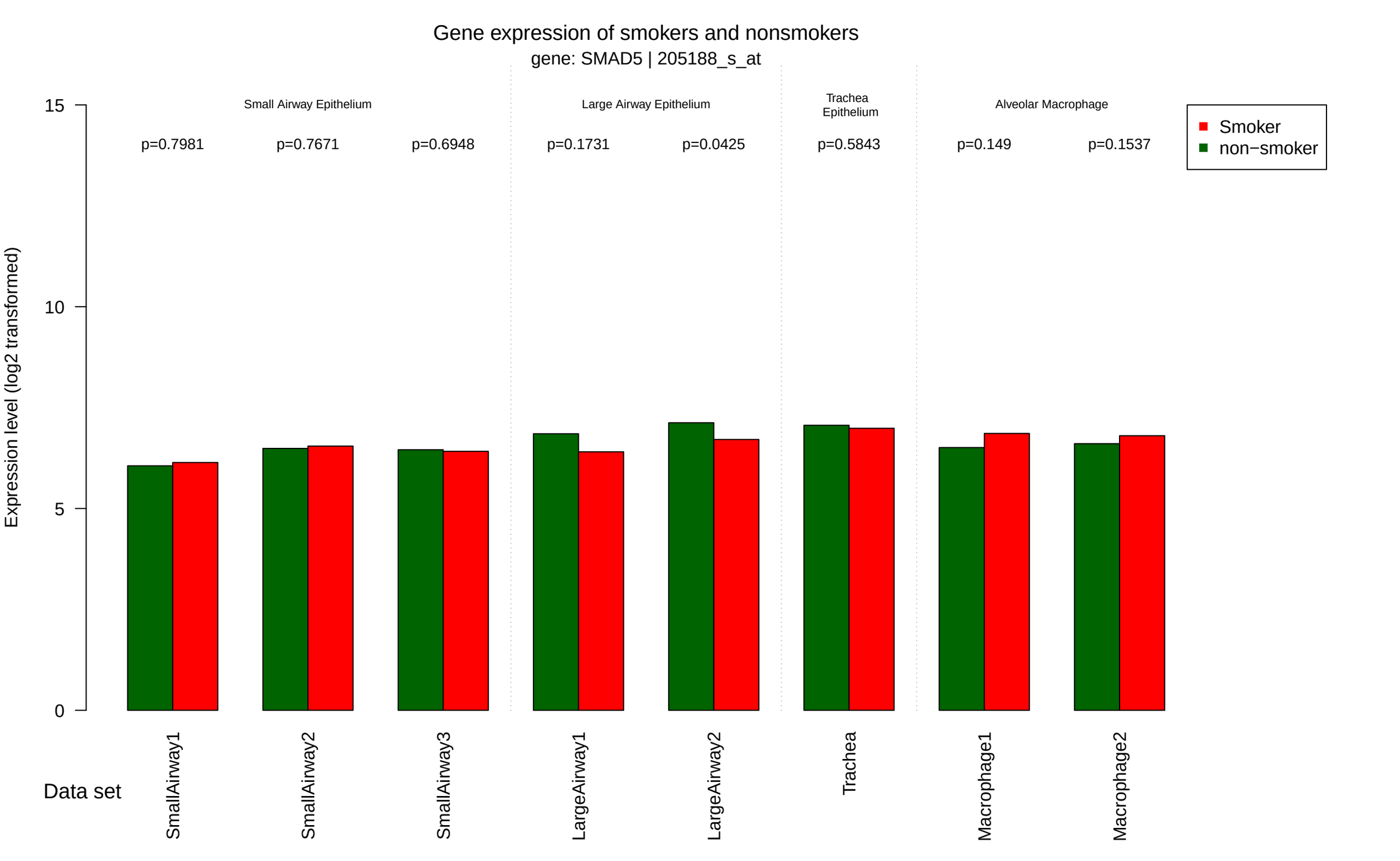
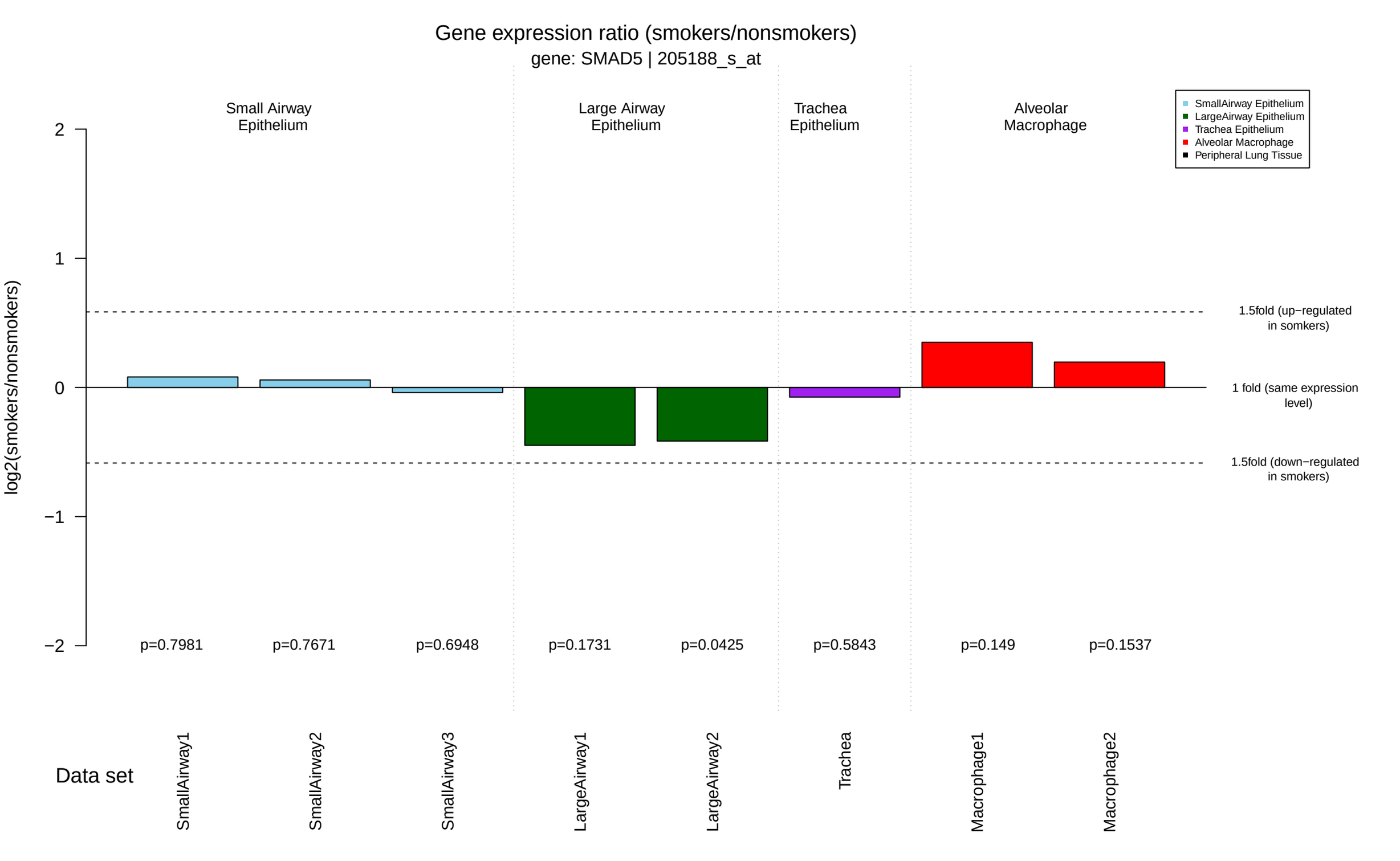
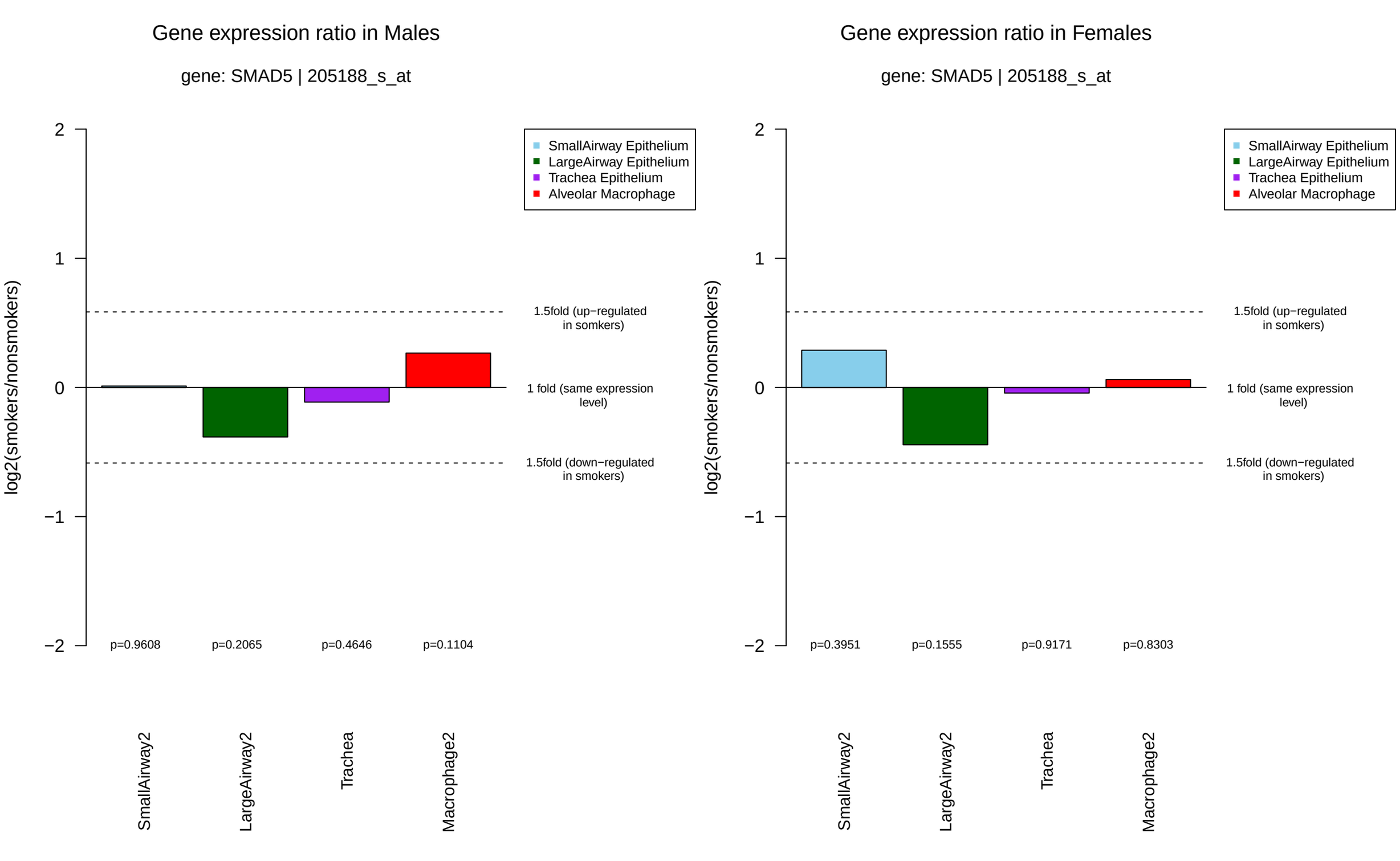
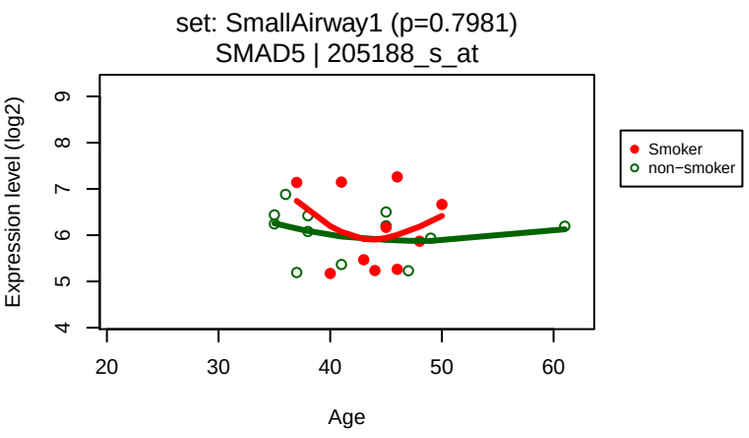
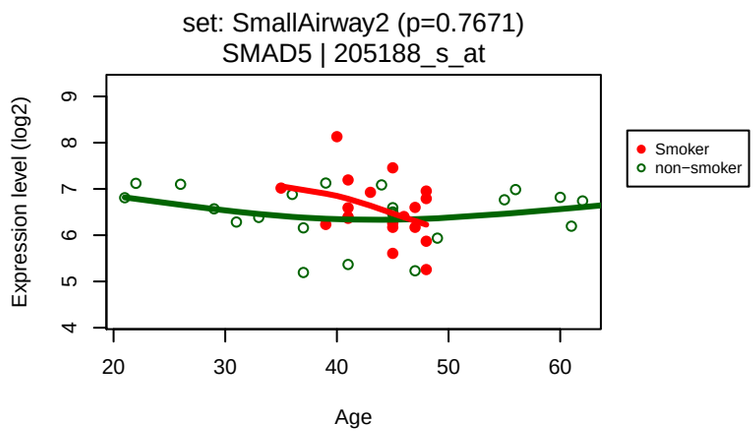
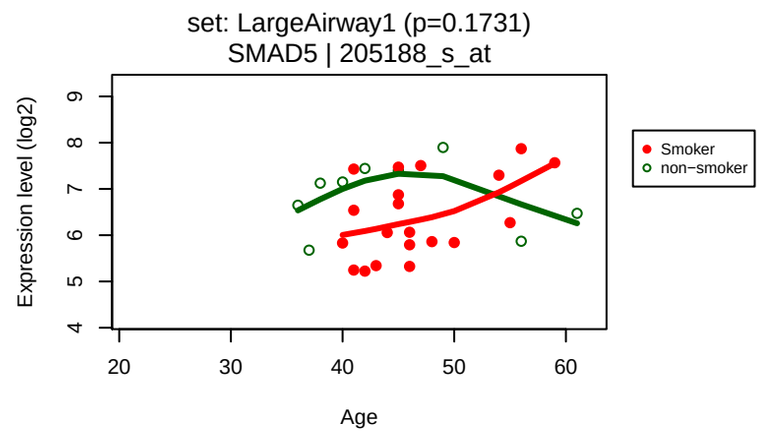

Gene expression of smokers and nonsmokers
gene: SMAD5 | 205188_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7981
p=0.7671
p=0.6948
p=0.1731
p=0.0425
p=0.5843
p=0.149
p=0.1537
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

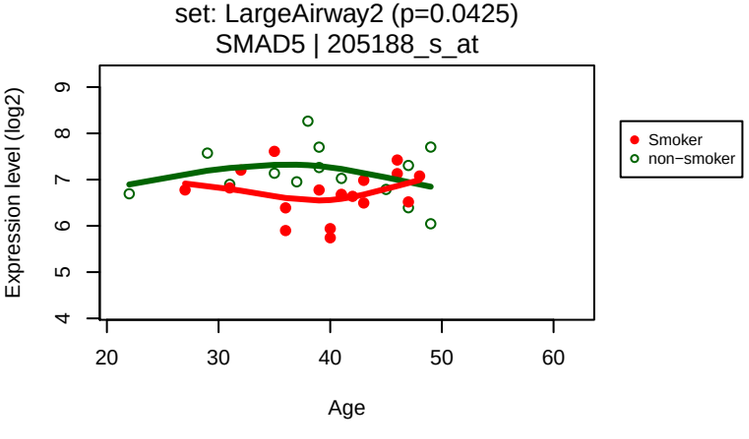

Gene expression ratio (smokers/nonsmokers)
gene: SMAD5 | 205188_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7981
p=0.7671
p=0.6948
p=0.1731
p=0.0425
p=0.5843
p=0.149
p=0.1537
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

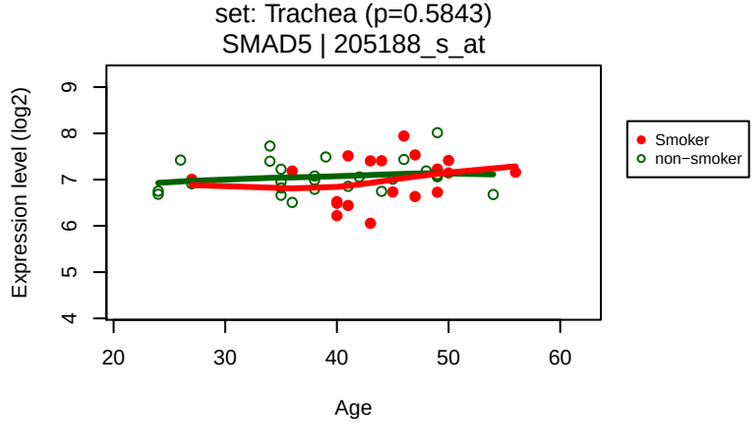

Gene expression ratio in Males
gene: SMAD5 | 205188_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9608
p=0.2065
p=0.4646
p=0.1104
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SMAD5 | 205188_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3951
p=0.1555
p=0.9171
p=0.8303
SmallAirway2
LargeAirway2
Trachea
Macrophage2

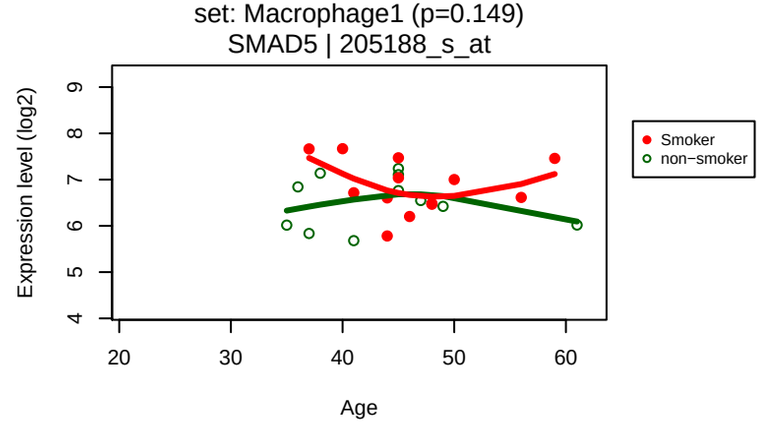

set: SmallAirway1 (p=0.7981)
SMAD5 | 205188_s_at
Expression level (log2)
Age
Smoker
non−smoker

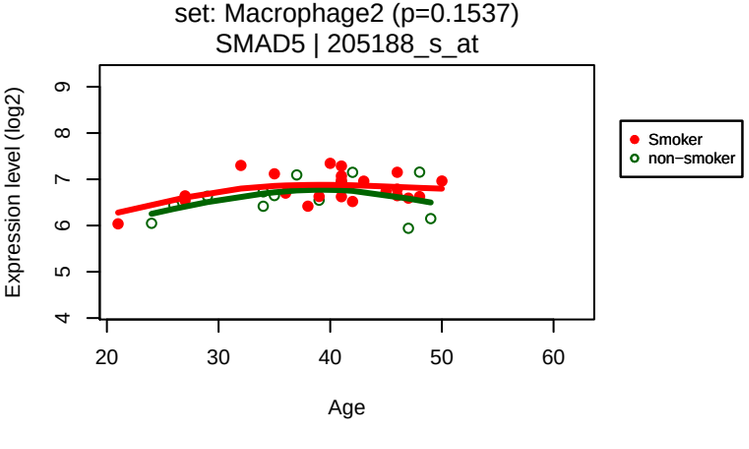

set: SmallAirway2 (p=0.7671)
SMAD5 | 205188_s_at
Expression level (log2)
Age
Smoker
non−smoker

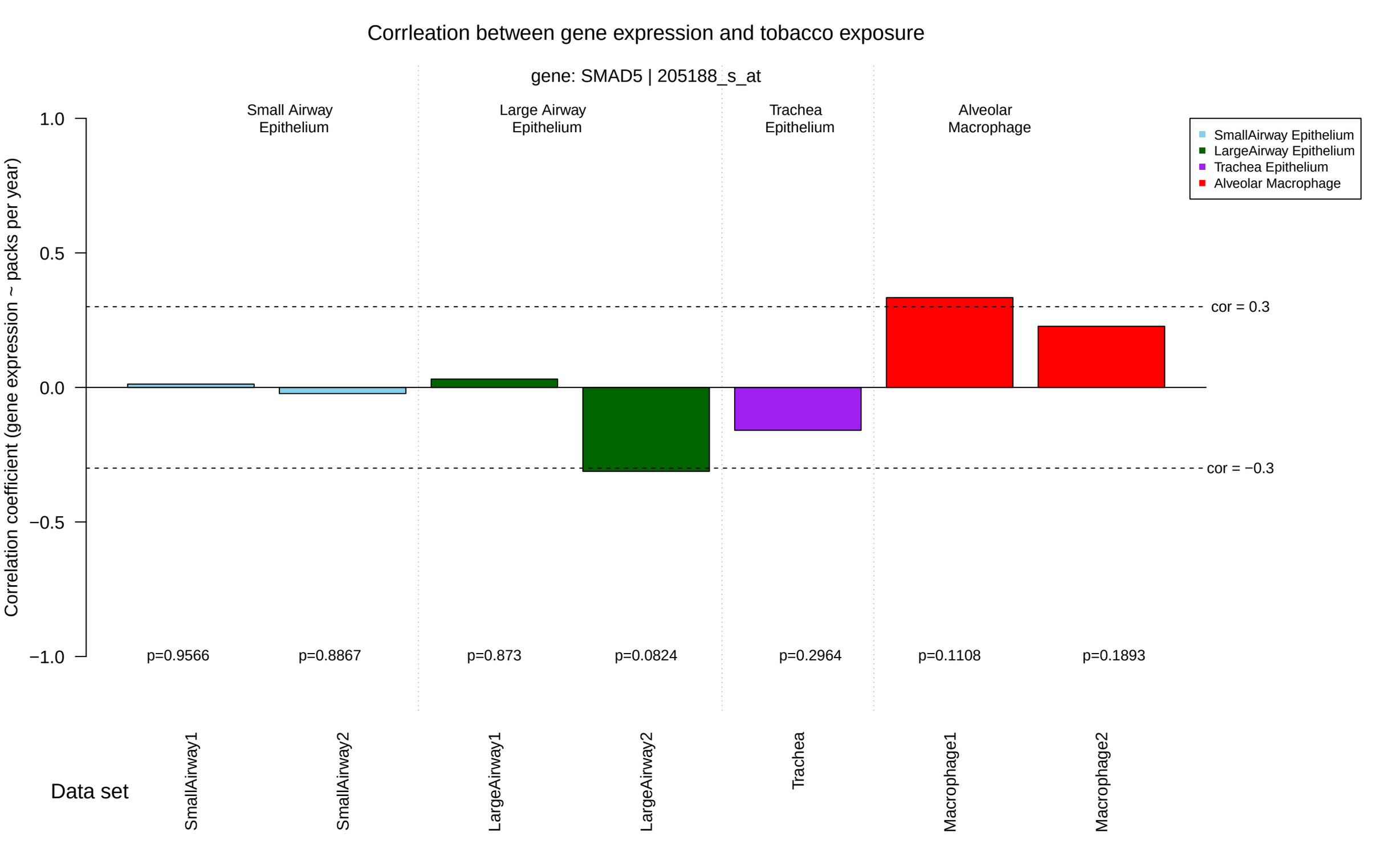

set: LargeAirway1 (p=0.1731)
SMAD5 | 205188_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0425)
SMAD5 | 205188_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5843)
SMAD5 | 205188_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.149)
SMAD5 | 205188_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1537)
SMAD5 | 205188_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SMAD5 | 205188_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9566
p=0.8867
p=0.873
p=0.0824
p=0.2964
p=0.1108
p=0.1893
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2